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ORIGINAL RESEARCH

Myoelectric Activation Pattern Changes in the Involved Limb of Individuals With Transtibial Amputation During Locomotor State Transitions

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Abstract

Objective: To investigate whether lower extremity muscle activation patterns differ in the strides leading to locomotive state transitions in the involved limb of individuals with transtibial amputation. It is hypothesized that all transitions elicit activation differences between strides as the subjects move toward the transition event.

Design: Single-sample, observational study.

Setting: University research center.

Participants: Volunteer sample of persons with unilateral transtibial amputation (N=9; mean age, 48.8±12.1y; mean height, 1.74±0.09m; mean weight, 86.1±24.7kg) were recruited by posting flyers in local prosthetics clinics.

Interventions: Not applicable.

Main Outcome Measures: Surface electromyography was used to measure muscle activation from 7 muscles of the involved limb. Subjects walked across 8 different terrain conditions transitioning from level-ground to ramp/stair locomotion and vice versa. Statistical Parametric Mapping analysis of variance ($\alpha<.05$) was used to assess muscle activation differences in the 3 pretransition gait cycles as subjects moved toward the terrain change.

Results: No muscle activation changes were observed in ramp transitions. All stair transitions elicited a myoelectric difference in at least 2 muscles. The transition from stair descent to level ground elicited change in the greatest number of muscles. Tibialis anterior activation was unchanged in all transitions.

Conclusions: Muscle activation differences were observed in the involved limb of individuals with transtibial amputation during stair transitions, suggesting that those patterns may be successfully used in transition detection algorithms. It remains unknown whether ramp transitions can be successfully identified pretransition using electromyography.

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Individuals with lower limb amputation face numerous challenges in daily activities. Asymmetric gait is common, propagating into secondary musculoskeletal ailments of the involved limb, contralateral limb, and lower back.[1] Active prostheses mimicking intact leg function have been suggested as a mechanism to reduce incidence of secondary pathologies.

Previous studies have used neuromuscular activity, in the form of electromyography, to develop prosthetic controllers.[2-6] Huang et al[3] developed classification algorithms for single-state, continuous locomotion. These algorithms may allow locomotion with appropriate kinematics on isolated terrain types. This provides an incomplete picture of locomotion because it neglects transitions between locomotion states occurring throughout the day.

Previous studies of level-ground,[7] ramp,[8-10] and stair[11-13] locomotion have established that these terrains require different kinematics. Kinematic differences have also been documented in individuals with transtibial amputation.[14,15] It is important to understand how these individuals transition between terrains, to

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better inform development of locomotion classification algorithms for the next generation of powered prostheses.

Recent work used mechanical sensor data to classify locomotion states and transitions.[16] Another method described transitions as a statistically separate class, suggesting it may be classified using steady-state locomotion approaches.[17] Few studies have examined myoelectric activation during the actual transition between locomotor states, using able-bodied subjects in a controlled laboratory environment.[18-21]

Little is known about muscle activation patterns in persons with transtibial amputation during locomotor transitions between level-ground, ramp, and stair terrains. The purpose of this study was to determine whether lower extremity muscle activation differences occur in the involved limb of individuals with transtibial amputation during locomotive state transitions, using their prescribed passive prostheses. It was hypothesized that all transition types would elicit activation differences.

Methods

Individuals with unilateral transtibial amputation were recruited using the following criteria: at least 1 year since amputation, no complicating musculoskeletal injuries since amputation, currently using a passive prosthesis, and ability to ambulate over level ground (LG), stairs, and ramps. Patients with dysvascular complication were excluded to minimize associated comorbidities. Subjects provided written informed consent prior to participation in the institutional review board—approved protocol.

Surface electromyographic data were collected from 7 muscles of the involved limb (fig 1). Passive surface electrodes (Ag/Ag-Cl) were placed on the rectus femoris (RF), vastus lateralis (VL), biceps femoris (BF), gluteus maximus (Gmax), and gluteus medius (Gmed) using common placement protocols.[22] These muscles were selected as representatives of the major muscle groups considered primary sagittal plane movers. Neonatal electrodes[a] were used in lieu of standard surface electrodes on muscles within the prosthetic socket (see fig 1D): tibialis anterior (TA) and medial gastrocnemius (MG). The difference in electrode type between within-socket and out-of-socket muscles should not affect study outcomes because of magnitude normalization, subsequently described in this section. Muscle belly location for the TA and MG were often compromised as a result of amputation. To identify the muscle belly location, subjects were asked to imagine flexing their toes up (dorsiflexion) and down (plantarflexion), while the investigator palpated the most contractile portion of the muscle. Electrode placement was confirmed with follow-up on contractions to ensure that signals were being recorded from residual muscle tissue. Local transmitters[b] were placed lateral to electrode placement sites. Cohesive flexible bandaging was used to secure wires and transmitters to reduce motion artifact. Footswitch insoles[b] were inserted into the subjects' shoes to collect the gait events of foot contact and toe off. Muscle activation data recorded from the noninvolved limb are not reported in this article because the focus is on changes within the involved limb and how those changes may inform design of classification approaches for prosthetic limbs.

Subjects completed 24 successful trials at self-selected normative walking pace. Trials were grouped into 8 different transition types: LG to ramp ascent, LG to ramp descent, ramp ascent to LG, ramp descent to LG, LG to stair ascent (SA), LG to stair descent (SD), SA to LG, and SD to LG. Trials that used the same type of transition, but in the opposite direction, (ie, LG ramp ascent and ramp descent to LG) were grouped and alternated to reduce fatigue. Subjects began each trial at a position which allowed completion of 4 strides before the transition stride to ensure continuous steady performance of the first locomotion state before transition. Transitions were completed on ramps with a grade of 5° and stairs with a height of 16.5cm and depth of 30.5cm. For each transition, subjects completed 3 successful trials. A successful trial was defined as transitioning with their involved leg and having completed the trial without any complications. The transitioning leg was defined as the first leg to perform a maneuver qualitatively different from the previous state. In most conditions, the first limb to land on the ensuing state was considered the transitioning limb. For example, for LG to SA, when the subject's involved limb contacted the first stair, it was defined as the transitioning limb. However, for SA to LG and SD to LG, the second limb to contact LG was considered the transitioning limb because this limb was the first to perform a different task. For example, the first foot to contact LG after a steady-state SA condition would effectively be ascending 1 more stair; however, the next limb would then swing up from the last stair surface, and change its pattern to extend forward into the flat terrain of LG.

Three gait cycles per trial were analyzed in this study (fig 2). Gait cycles were defined from heel-strike to heel-strike of the involved leg. The gait cycle beginning 3 heel-strikes before transition, 2 heel-strikes before transition, and 1 heel-strike before transition were analyzed. Gait cycles were partitioned into stance and swing phase at toe-off. The electromyographic and foot switch data were sampled at 1500Hz. Gait cycles were ensemble averaged by up-sampling trials, using spline interpolation, to the largest trial length per transition for each subject. Relative timing of stance and swing (percent of gait cycle) for stair locomotion trials were observed to be within 1% in LG trials. The electromyographic signals were then bandpass filtered (3−500Hz), full wave rectified, and smoothed using a fourth-order Butterworth low-pass filter (cutoff frequency, 6Hz). Each trial was then amplitude normalized to the maximum amplitude of that trial; considered to be 100% activation. This allowed transition to a new locomotion state to show relative changes, with respect to the maximum signal of the starting condition. These signal processing steps were adopted from previously published methods.[23,24]

Statistical analysis

One-dimensional Statistical Parametric Mapping (SPM) analysis of variance ($\alpha<.05$) was used.[25,26] Figure 3 shows representative graphs of the SPM analysis of variance results. The bottom panel in figure 3A illustrates the ensemble average of electromyographic

List of abbreviations:

BF	**biceps femoris**
Gmax	**gluteus maximus**
Gmed	**gluteus medius**
LG	**level ground**
MG	**medial gastrocnemius**
RF	**rectus femoris**
SA	**stair ascent**
SD	**stair descent**
SPM	**Statistical Parametric Mapping**
TA	**tibialis anterior**
VL	**vastus lateralis**

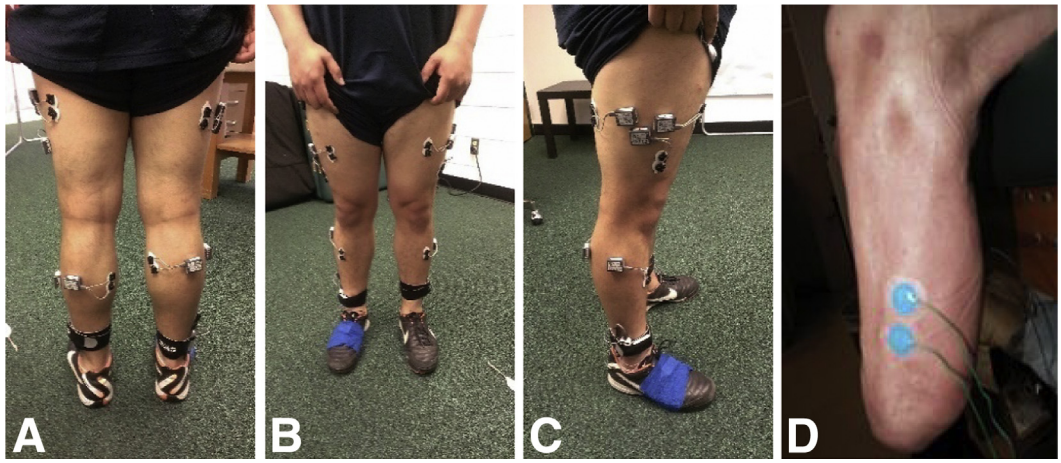

Fig 1 Electromyographic electrode, transmitter, and footswitch setup: (A) posterior, (B) anterior, and (C) lateral views of able-bodied subject setup. (D) Representative setup of low-profile neonatal electromyographic electrodes on residual limb for within-socket musculature.

activation for 3 gait cycles preceding transitions for 1 muscle across all subjects. The top panel in figure 3A illustrates the running f value of the SPM analysis of variance. The f critical (determined by the α level and standard F distribution tables) is designated by the horizontal dotted line. If the f value exceeds the f critical, the data at that point are considered significantly different. Using SPM to analyze the time series is similar to using a scalar 2-sample t test, with the addition of an internal statistic calculated to account for any covariance between the 2 time series. This approach allows comparisons to be made in vector field reference, rather than scalar, as is the case with traditional 2-sample t test approaches.[25-27] Cycle differences leading up to the transition were analyzed. Follow-up, pairwise analyses were also conducted using SPM t tests, with a Bonferroni-corrected α level ($\alpha<.017$) to determine differences by gait cycle as subjects approached the transition (fig 3B). Using data from a previous study,[24] the estimated sample size was 10, based on a power analysis using a mean detectable difference of 28% activation and SD of 13.3% relative activation, with a 2-sided α level of .05, resulting in power of 87%.

Results

Nine individuals with unilateral transtibial amputation (mean age, 48.8±12.1y; mean height, 1.74±0.09m; mean weight, 86.1±24.7kg) were recruited for this study. Average time post-amputation was 9.3±9.3 years (range, 1−30y). Average residual limb length was 15.5±2.0cm (range, 13.5−20cm). Cause of amputation included trauma (n=6), Berger disease (n=1), infection (n=1), and cancer (n=1). Each subject wore their prescribed prosthetic foot and their prescribed socket system. Patient demographics are provided in table 1. The actual sample (N=9) reduced the a priori estimate of statistical power (n=10) from 87% to 82%.

A graphical depiction of the SPM analysis results is presented in figure 4. Ramp transitions did not elicit a significant change in any musculature and are therefore not included in the presentation of results. Within stair transitions, 6 muscles showed activation differences with TA being the exception. Individually, each muscle exhibited significant activation differences in 2 of the 4 stair transitions. Follow-up pairwise comparison showed that the 3 heel-strikes before transition/1 heel-strike before transition and 2 heel-strikes before transition/1 heel-strike before transition comparisons were significant.

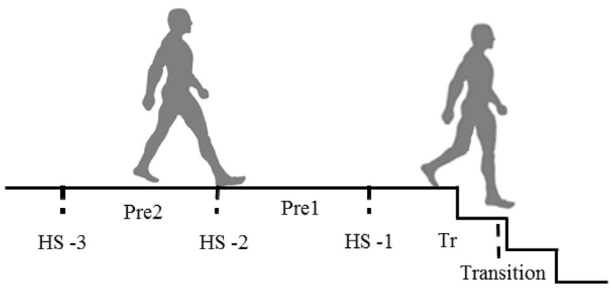

Fig 2 Gait cycle definitions with heel-strike designations from LG to SD. Gait cycles are defined from heel-strike to heel-strike of the right leg. Abbreviations: HS, heel strike; Pre1, 2 heel-strikes before transition; Pre2, 3 heel-strikes before transition; Tr, 1 heel-strike before transition.

LG to SA

The MG and Gmed were found to have different activation patterns in the LG to SA transition. Increased Gmed activation was found in late stance through early swing (58%−66% of gait cycle). Late in the swing phase, a decrease in Gmed activation was observed from 89% to 100%. A brief decrease in MG activation was observed from 87% to 89%.

LG to SD

During the LG to SD transition, RF and VL were observed to have activation differences in the swing and stance phases, respectively. The VL had increased activation from 43% to 54%, whereas RF showed decreased activation from 67% to 70%.

SA to LG

The swing phase showed activation differences in both BF and Gmax during the SA to LG transition. The BF activation was decreased during the swing phase (71%−84%). Late in the swing phase, Gmax activation increased (95%−100%).

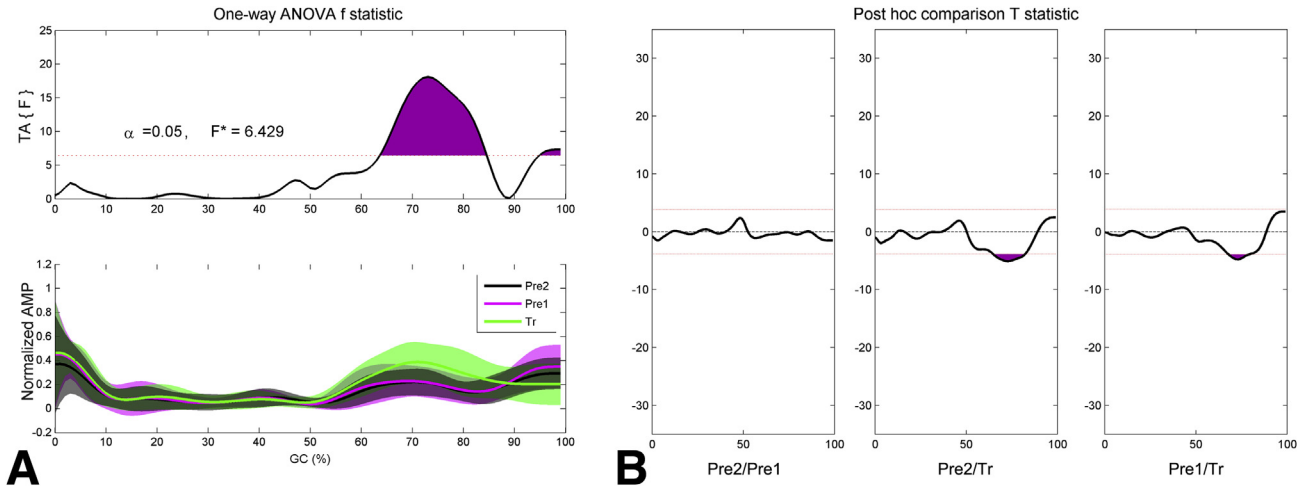

Fig 3 SPM analysis of variance results. (A) The bottom panel shows the ensemble-averaged electromyographic activation patterns for Pre2, Pre1, and Tr of a single transition for all subjects. Shaded regions indicate ±1 SD. The top panel shows the running f value, with the f critical shown as the dotted line. (B) Post hoc pairwise comparison of Pre2/Pre1, Pre2/Tr, and Pre1/Tr, with the running T statistic and the T critical band shown as the dotted band. In this example, the initial 1-way analysis of variance indicated significant effect, for which the post hoc comparison revealed significant differences between Pre2 and Tr, and Pre1 and Tr. Abbreviations: AMP, amplitude; GC, gait cycle; Pre1, 2 heel-strikes before transition; Pre2, 3 heel-strikes before transition; Tr, 1 heel-strike before transition.

SD to LG

The SD to LG transition showed altered activation patterns in 5 of the 7 assessed muscles. Greater levels of activation primarily occurred around toe-off/early swing phase. The RF, VL, BF, and Gmax increased activation primarily from 52% to 74%. MG showed increased activation from 81% to 92% of the gait cycle.

Discussion

The aim of this study was to determine if there were myoelectric differences in the musculature of the involved limb in a sample of individuals with transtibial amputation during transitions between locomotor states, using their prescribed passive prostheses. It was hypothesized that all transitions, to and from ramp and stair locomotion, would yield significant differences. Significant differences were observed in 4 transitions and 6 muscles. None of the ramp transitions showed myoelectric differences. This corroborates findings in able-bodied studies where myoelectric differences were observed, but only at higher grades.[21,28] The ramp grade used in this study (5°) is in the low to mid-range of the grades assessed in previous studies. This is the standard set by the Americans with Disabilities Act, representing what would be a common setting in daily life.

LG to SA

The Gmed showed the most prominent difference in this transition. Just before toe-off, Gmed activation increased through early swing. This activation change could be a result of 2 mechanisms. Persons with lower limb amputation often have asymmetric gait, which can lead to decreased stability.[29] Increased pelvic stability, via Gmed activation would be necessary during this phase. Additionally, in conjunction with whole-body sway at toe-off, slight hip abduction of the involved limb would be necessary to maintain ordinary swing kinematics.

The late swing deactivation of MG and Gmed begin simultaneously. Because the MG attaches on the medial femoral condyle,

Table 1 Research participant demographics

Subject No.	Sex	Age (y)	Time Postamputation (y)	Etiology	Limb Length (cm)	Foot Type	Suspension
1	F	47	11	MVC	14.0	DER	Vacuum
2	M	32	2	Work injury	20.0	DER	Vacuum
3	M	53	1	Berger disease	14.2	DER	Pin-lock
4	M	63	6	Accident/infection	15.4	DER	Pin-lock
5	M	54	30	MVC	17.0	DER	Pin-lock
6	M	58	14	MVC	13.5	DER	Pin-lock
7	F	62	5	MVC	15.2	DER	Pin-lock
8	M	34	1	Cancer	15.8	DER	Pin-lock
9	M	36	12	MVC	14.2	DER	Pin-lock

NOTE. All subjects were K-level III.
Abbreviations: DER, dynamic elastic response; F, female; M, male; MVC, motor vehicle collision.

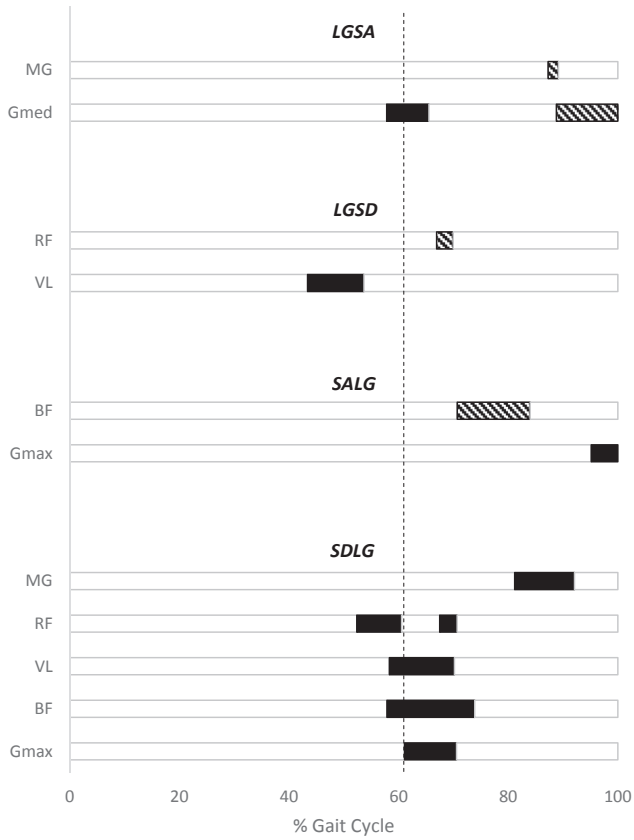

Fig 4 Significant activation differences as determined by SPM analysis of variance. Black shaded areas represent significant increases in activation. Striped areas represent decreases in activation during the respective portions of the gait cycle. The vertical dashed line represents toe-off between the stance and swing phases. Abbreviations: LGSA, LG to SA transition; LGSD, LG to SD transition; SALG, SA to LG transition; SDLG, SD to LG transition.

brief deactivation could allow subjects to reduce their knee flexor moment to allow angular momentum of the lower limb to passively move the leg into appropriate stair kinematics. In combination with MG, Gmed deactivation also suggests utilization of momentum generated earlier in swing. Increased activation in early swing may slightly abduct the leg; therefore, late swing deactivation may correct earlier kinematic differences as the shank progresses toward the stair.

LG to SD

Only quadriceps muscles showed activation differences in the LG to SD transition. The RF and VL acted independently from one another. This suggests that dual roles of the RF as a hip flexor and knee extensor were in effect. Late in the stance phase, the increased VL activation suggests that the quadriceps were providing a propulsive knee extensor moment. However, during this portion of stance, subjects should be extending their hip, requiring RF coactivation.

Early in the swing phase, the RF was observed to briefly decrease activation. The dual role of the RF can also explain this decrease in activation. To transition from LG to SD locomotion, the leg needs to descend toward the stair, requiring hip and knee extension. Acting eccentrically at the hip, the RF would need to reduce activation to allow passive knee extension. Eccentric hip activation has been reported in LG locomotion; however, passive knee extension has not.[30]

SA to LG

Myoelectric changes were observed in the BF and Gmax during SA to LG, during the swing phase. The deactivation of the BF in mid-swing is interesting because it may be a result of prosthesis compensation. While ascending stairs, the BF is activated to sustain and carry the prosthesis over the ensuing stair.[31] This is necessary because the TA no longer controls dorsiflexion to adjust for toe clearance. Therefore, individuals with transtibial amputation will compensate by flexing the knee more to gain sufficient clearance height. On the final step before transition, the clearance height is decreased because there is only a single step remaining before LG.

Increased Gmax activation may be preemptive for the upcoming gait cycle. In SA locomotion, initial stance requires Gmax to stabilize the hip joint via a hip extensor moment. When transitioning to LG locomotion, the hip extensor moment is increased to accommodate the need to move forward.

SD to LG

Most observed changes during SD to LG occurred around toe-off. The RF, BF, VL, and Gmax showed increased activation, with temporal onset occurring in that order. Increased activation in multiple muscles suggests that the period between stance and swing is complicated for these patients in SD to LG transitions. A previous study of ramp locomotion suggested that increased muscle coactivation was indicative of an upcoming transition.[19] The combined activation effect discovered in this transition may serve as a compensatory mechanism to ensure successful ground clearance by the prosthesis during the initial stages of swing phase. This may be accomplished through a slight delay in hip flexion with Gmax activation, sustained knee flexion with BF and VL cocontraction, and rapid hip flexion from the RF to bring the prosthesis under the body.

The interplay between the RF and Gmax is intriguing. The RF returns to a normative level, whereas Gmax activation remains elevated. This suggests a coordinated effort to maintain an appropriate level of hip joint cocontraction to position the thigh. Because the thigh muscles are the most distal noncompromised musculature in persons with transtibial amputation, it may be simpler to manipulate and coordinate these muscles than the shank musculature.

Study limitations

The gait of individuals with transtibial amputation is varied and depends on factors such as length of residual limb, amount of gait rehabilitation, secondary musculoskeletal pathologies, and motivation for gait improvement. In this study, all subjects were able to successfully complete each transition and were not given assistance. However, individual subject confidence varied and was evident in how they approached the transitions. Additionally, patients with dysvascular complication were excluded in this study to minimize associated comorbidities. This limits the general application of the current study's findings to those patients without dysvascular etiology.

The research protocol for this study required subjects to begin walking from a position allowing 4 complete strides prior to the transition stride. This approach allowed for consistent performance by each subject. Real-world environments would present more variable scenarios. An individual may experience a variety of terrains prior to a transition stride. Individuals will encounter slopes of varying degree in their everyday environment. These variations may affect our system's ability to detect changes in lower extremity muscle activation patterns. These findings are limited to the Americans with Disabilities Act standard slope.

This study used electromyographic data from muscles representing the muscle groups considered to be primary movers in the sagittal plane. However, other muscle groups can also play an important role in the gait of individuals with lower limb loss (eg, hip adductors). Additional differences may have been observed if these muscle groups were included.

The shape and orientation of the MG and TA after transtibial amputation are different from the original anatomic arrangement. Surgical techniques can vary, with the posterior myocutaneous flap technique, where the soleus muscle and the gastrocnemius tendon are sutured to the anterior tibia, being common.[32] The variety of surgery outcomes can result in variable muscular alignments which are unique to each individual. This poses a unique challenge to measurement of the electromyographic signal. However, our observation of significant differences between pretransition and transition cycles suggests invariant patterns of transition which the residual muscles are able to retain. Additionally, the electromyographic magnitude normalization technique used in this study (dynamic signals normalized to the maximum signal during a condition) allows for within-condition assessment of changes, but may not allow for between-condition assessments. This limits application of these findings to a series of within-condition classifiers, reducing the possibility of deploying a universal classifier.

These findings suggest that transition detection would need to include data from muscles superior to the socket. This limits eventual application of these findings because it may be prohibitive to place electrodes outside of the socket. Less obtrusive kinematic and kinetic sensors could be integrated to reduce this limitation. These combinations have been used successfully in steady-state locomotion classification in samples of individuals with lower limb amputation,[3,33] and transition classification for able-bodied subjects.[18,34]

The findings in this study provide insight into which phases of gait could be used for enhanced validity and efficiency in classification algorithms. The lack of significant differences in ramp transitions begs the question of whether ramp transition classification is important or whether those transition types must use some signal other than electromyography.

Conclusions

Myoelectric activation of lower extremity musculature was altered in the involved limb of persons with nondysvascular unilateral transtibial amputation using prescribed passive prostheses when performing transitions involving stair locomotion, but not during transitions involving Americans with Disabilities Act standard ramps. Therefore, electromyography as sensor input for transition classification may only be useful for stair locomotion. Additional research is needed to determine the viability of a classifier for stair transitions in combination with alternative strategies for ramp transitions.

Suppliers

a. BlueSensor NF; Ambu A/S.
b. Telemyo DTS; Noraxon USA.

Keywords

Amputation; Electromyography; Locomotion; Lower extremity; Rehabilitation

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References

1. Gailey R, Allen K, Castles J, Kucharik J, Roeder M. Review of secondary physical conditions associated with lower-limb amputation and long-term prosthesis use. J Rehabil Res Dev 2008;45:15-29.
2. Farmer S, Silver-Thorn S, Voglewede P, Beardsley SA. Within-socket myoelectric prediction of continuous ankle kinematics for control of a powered transtibial prosthesis. J Neural Eng 2014;11:056027.
3. Huang H, Zhang F, Hargrove LJ, Dou Z, Rogers DR, Englehart KB. Continuous locomotion-mode identification for prosthetic legs based on neuromuscular-mechanical fusion. IEEE Trans Biomed Eng 2011; 58:2867-75.
4. Huang S, Wensman JP, Ferris DP. An experimental powered lower limb prosthesis using proportional myoelectric control. J Med Device 2014;8.
5. Ohnishi K, Weir RF, Kuiken TA. Neural machine interfaces for controlling multifunctional powered upper-limb prostheses. Expert Rev Med Devices 2007;4:43-53.
6. Parker P, Englehart K, Hudgins B. Myoelectric signal processing for control of powered limb prostheses. J Electromyogr Kinesiol 2006;16:541-8.
7. Winter DA. Kinematic and kinetic patterns in human gait: variability and compensating effects. Hum Mov Sci 1984;3:51-76.
8. Lay AN, Hass CJ, Gregor RJ. The effects of sloped surfaces on locomotion: backward walking as a perturbation. J Biomech 2007;40: 3050-5.
9. McIntosh AS, Beatty KT, Dwan LN, Vickers DR. Gait dynamics on an inclined walkway. J Biomech 2006;39:2491-502.
10. Redfern MS, DiPasquale J. Biomechanics of descending stairs. Gait Posture 1997;6:119-25.
11. Andriacchi TP, Andersson GB, Fermier RW, Stern D, Galante JO. A study of lower-limb mechanics during stair-climbing. J Bone Joint Surg Am 1980;62:749-57.
12. McFadyen BJ, Winter DA. An integrated biomechanical analysis of normal stair ascent and descent. J Biomech 1988;21:733-44.
13. Reiner R, Rabuffetti M, Frigo C. Biomechanical analysis of stair ambulation in lower limb amputees. Gait Posture 2002;25:267-78.
14. Schmalz T, Blumentritt S, Marx B. Biomechanical analysis of stair ambulation in lower limb amputees. Gait Posture 2007;25:267-78.
15. Segal AD, Orendurff MS, Czerniecki JM, Schoen J, Klute GK. Comparison of transtibial amputee and non-amputee biomechanics during a common turning task. Gait Posture 2011;33:41-7.
16. Young AJ, Simon AM, Hargrove LJ. A training method for locomotion mode prediction using powered lower limb prostheses. IEEE Trans Neural Syst Rehabil Eng 2014;22:671-7.
17. Young AJ, Hargrove LJ. A classification method for user-independent intent recognition for transfemoral amputees using powered lower limb prostheses. IEEE Trans Neural Syst Rehabil Eng 2016;24:217-25.

18. Peng J, Fey NP, Kuiken TA, Hargrove LJ. Anticipatory kinematics and muscle activity preceding transitions from level-ground walking to stair ascent and descent. J Biomech 2016;49:528-36.
19. Gottschall JS, Nichols TR. Neuromuscular strategies for the transitions between level and hill surfaces during walking. Philos Trans R Soc Lond B Biol Sci 2011;366:1565-79.
20. Sheehan RC, Gottschall JS. Stair walking transitions are an anticipation of the next stride. J Electromyogr Kinesiol 2011;21:533-41.
21. Sheehan RC, Gottschall JS. Walking strategies change with distance from hill transition and scale with hill angle. J Appl Biomech 2012;28:738-45.
22. Delagi EF, Perotto A. Anatomic guide for the electromyographer—the limbs. 2nd ed. Springfield: Thomas; 1980.
23. Joshi D, Nakamura BH, Hahn ME. High energy spectrogram with integrated prior knowledge for EMG-based locomotion classification. Med Eng Phys 2015;37:518-24.
24. Seyedali M, Czerniecki JM, Morgenroth DC, Hahn ME. Co-contraction patterns of trans-tibial amputee ankle and knee musculature during gait. J Neuroeng Rehabil 2012;9:29.
25. Pataky TC, Robinson MA, Vanrenterghem J. Vector field statistical analysis of kinematic and force trajectories. J Biomech 2013;46:2394-401.
26. Robinson MA, Vanrenterghem J, Pataky TC. Statistical parametric mapping (SPM) for alpha-based statistical analyses of multi-muscle EMG time-series. J Electromyogr Kinesiol 2015;25:14-9.
27. Cao J, Worsley KJ. The detection of local shape changes via the geometry of Hotelling's T2 fields. Ann Stat 1999;27:925-42.
28. Franz JR, Kram R. The effects of grade and speed on leg muscle activations during walking. Gait Posture 2012;35:143-7.
29. Adamczyk PG, Kuo AD. Mechanisms of gait asymmetry due to push-off deficiency in unilateral amputees. IEEE Trans Neural Syst Rehabil Eng 2015;23:776-85.
30. Lyons K, Perry J, Gronley JK, Barnes L, Antonelli D. Timing and relative intensity of hip extensor and abductor muscle action during level and stair ambulation. An EMG study. Phys Ther 1983;63:1597-605.
31. Benedetti MG, Agostini V, Knaflitz M, Gasparroni V, Boschi M, Piperno R. Self-reported gait unsteadiness in mildly impaired neurological patients: an objective assessment through statistical gait analysis. J Neuroeng Rehabil 2012;9:64.
32. Brown BJ, Iorio ML, Klement M, et al. Outcomes after 294 transtibial amputations with the posterior myocutaneous flap. Int J Low Extrem Wounds 2014;13:33-40.
33. Young AJ, Kuiken TA, Hargrove LJ. Analysis of using EMG and mechanical sensors to enhance intent recognition in powered lower limb prostheses. J Neural Eng 2014;11:056021.
34. Joshi D, Hahn ME. Terrain and direction classification of locomotion transitions using neuromuscular and mechanical input. Ann Biomed Eng 2016;44:1275-84.